

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273423

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

MIYASAKA; Mika et al.

ELECTRON GUN AND ELECTRON BEAM WRITING APPARATUS

Abstract

An electron gun includes a cathode to emit electron beams, an anode configured to include a surface which faces the cathode and in which there are formed the first opening for passing the electron beams from the cathode and at least one second opening at a position different from that of the first opening and in the same surface as the first opening, and maintained to be a relatively positive potential with respect to a cathode potential, a limiting aperture substrate at the downstream side of the anode with respect to an advancing direction of the electron beams, formed with the third opening for passing the electron beams and limiting passage of a portion of the electron beams, and a Wehnelt electrode between the cathode and the anode, applied with a relatively negative potential with respect to an anode potential.

Inventors: MIYASAKA; Mika (Kawasaki-shi, JP), MIYAMOTO; Nobuo (Yokohama-shi, JP)

Applicant: NuFlare Technology, Inc. (Yokohama-shi, JP)

Family ID: 1000008434878

Assignee: NuFlare Technology, Inc. (Yokohama-shi, JP)

Appl. No.: 19/024642

Filed: January 16, 2025

Foreign Application Priority Data

JP	2024-028134	Feb. 28, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: H01J37/09 (20060101); H01J37/063 (20060101); H01J37/07 (20060101); H01J37/10 (20060101); H01J37/317 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from prior Japanese Patent Application No. 2024-028134 filed on Feb. 28, 2024 in Japan, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] Embodiments of the present invention relate to an electron gun and an electron beam writing apparatus, and, for example, to an electron source installed in a multiple beam writing apparatus.

Description of Related Art

[0003] The lithography technique which advances miniaturization of semiconductor devices is extremely important as a unique process whereby patterns are formed in semiconductor manufacturing. In recent years, with high integration of LSI, the line width (critical dimension) necessary for semiconductor device circuits is becoming increasingly narrower year by year. The electron beam writing technique, which intrinsically has excellent resolution, is used for writing or “drawing” patterns on a wafer and the like with electron beams.

[0004] For example, as a known example of employing the electron beam writing technique, there is a writing apparatus using multiple beams. Since writing with multiple beams can apply a lot of beams at a time, the writing throughput can be greatly increased compared to writing with a single electron beam. For example, a writing apparatus employing the multiple-beam system forms multiple beams by letting an electron beam emitted from an electron gun pass through a mask having a plurality of holes, performs blanking control for each beam, reduces each unblocked beam to generate a reduced mask image by an optical system, and deflects, by a deflector, a reduced beam to be applied to a desired position on a target object or “sample”.

[0005] In the electron gun which emits electron beams, after the electron beams pass through the anode, secondary electrons or reflected electrons are generated due to that a part of the electron beams collides with an aperture substrate, resulting in generation of cations (or “positive ions”) from gas molecules in the atmosphere. Then, if the cations flow backwards (reversely) and collide with the surface of the cathode (emitter), crystals may be damaged, which poses a problem of incapable of securing electron beams with a sufficient current distribution. It becomes necessary to exchange the cathode from which electron beams having a sufficient current distribution are not emitted/acquired.

[0006] With regard to an electron gun mounted in an X-ray device, there is disclosed a technique which forms a plurality of exhaust paths in the side surface of a tubular electron passage hole arranged at the cathode side from the anode electrode surface in order to decrease the pressure in the electron passage hole through the exhaust paths, thereby reducing generation of cations (e.g., refer to Japanese Patent Application Laid-open (JP-A) No. 2012-256441). However, this poses a problem when a tubular member of a positive potential with respect to the cathode exists at the cathode side from the anode electrode surface, the risk of electric discharge increases.

BRIEF SUMMARY OF THE INVENTION

[0007] According to one aspect of the present invention, an electron gun includes [0008] a cathode

configured to emit an electron beam; [0009] an anode electrode configured to include a surface which faces the cathode, and in which there are formed a first opening for passing the electron beam emitted from the cathode, and at least one second opening at a position different from a position of the first opening, the at least one second opening being formed in a same surface as the first opening, and maintained at a relatively positive potential with respect to a potential of the cathode; [0010] a limiting aperture substrate arranged at a downstream side of the anode electrode with respect to an advancing direction of the electron beam, configured to be formed with a third opening for passing the electron beam, and limit passage of a portion of the electron beam; and [0011] a Wehnelt electrode arranged between the cathode and the anode electrode, configured to be applied with a relatively negative potential with respect to a potential of the anode electrode. [0012] According to another aspect of the present invention, an electron beam writing apparatus includes [0013] a stage configured to mount thereon a target object; [0014] an electron gun configured to include [0015] a cathode which emits an electron beam, [0016] an anode electrode configured to include a surface which faces the cathode, and in which there are formed a first opening for passing the electron beam emitted from the cathode, and at least one second opening at a position different from a position of the first opening, the at least one second opening being formed in a same surface as the first opening, and maintained at a relatively positive potential with respect to a potential of the cathode, [0017] a limiting aperture substrate arranged at a downstream side of the anode electrode with respect to an advancing direction of the electron beam, configured to be formed with a third opening for passing the electron beam, and limit passage of a portion of the electron beam, and [0018] a Wehnelt electrode arranged between the cathode and the anode electrode, configured to be applied with a relatively negative potential with respect to a potential of the anode electrode; and [0019] an electron optical system configured to lead the electron beam emitted from the electron gun to the target object.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIG. 1 is an illustration showing a schematic configuration diagram of a writing apparatus according to a first embodiment;

[0021] FIG. 2 is an illustration showing an example of a circuit configuration of an electron gun and a high-voltage power supply circuit according to the first embodiment;

[0022] FIG. 3 is a top view showing an example of a configuration of an anode electrode according to the first embodiment;

[0023] FIG. 4 is an illustration showing an example of an internal configuration of an electron gun according to a comparative example of the first embodiment;

[0024] FIG. 5 is an illustration showing an example of an internal configuration of an electron gun according to the first embodiment;

[0025] FIG. 6 is an illustration showing an example of a result of a simulation where cations flow backwards according to a comparative example of the first embodiment;

[0026] FIG. 7 is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening at a position away from an electron beam passage hole by a distance y_1 according to the first embodiment;

[0027] FIG. 8 is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening at a position away from an electron beam passage hole by a distance y_2 according to the first embodiment;

[0028] FIG. 9 is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening at a position away from an electron beam passage hole by a distance y_3 according to the first embodiment;

[0029] FIG. **10** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening with a diameter size **D21**;

[0030] FIG. **11** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening with a diameter size **D22**;

[0031] FIG. **12** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening with a diameter size **D23**;

[0032] FIG. **13** is an illustration showing a positional relationship in an electron gun according to the first embodiment;

[0033] FIG. **14** is a graph showing an example of a relationship between an optimal distance from a passage hole of an anode to an opening and a distance between the anode and a Wehnelt according to the first embodiment;

[0034] FIG. **15** is a table showing an example of a relationship between an optimal distance from a passage hole of an anode to an opening and a distance between the anode and a Wehnelt according to the first embodiment;

[0035] FIG. **16** is an illustration showing an example of an internal configuration of an electron gun according to a modified example 1 of the first embodiment;

[0036] FIG. **17** is an illustration showing an example of an internal configuration of an electron gun according to a modified example 2 of the first embodiment;

[0037] FIG. **18** is an illustration showing another example of a return portion at the anode backside according to the modified example 2 of the first embodiment;

[0038] FIG. **19** is a conceptual diagram showing a configuration of a shaping aperture array substrate according to the first embodiment;

[0039] FIG. **20** is a sectional view showing a configuration of a blanking aperture array mechanism according to the first embodiment;

[0040] FIG. **21** is a conceptual diagram showing an example of writing operations according to the first embodiment;

[0041] FIG. **22** is an illustration showing an example of an irradiation region of multiple beams and a writing target pixel according to the first embodiment; and

[0042] FIG. **23** is an illustration explaining an example of a multi-beam writing operation according to the first embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0043] Embodiments of the present invention provide an apparatus capable of preventing that cations (or “positive ions”) generated due to collision of electrons having passed through the opening of the anode electrode flow backwards (reversely) to the cathode.

[0044] Embodiments of the present invention describe a configuration which uses multiple electron beams. However, the electron beam is not limited to multiple beams, and a single beam may also be used.

First Embodiment

[0045] FIG. **1** is an illustration showing a schematic configuration diagram of a writing or “drawing” apparatus according to a first embodiment. As shown in FIG. **1**, a writing apparatus **100** includes a writing mechanism **150** and a control system circuit **160**. The writing apparatus **100** is an example of a multiple electron beam writing apparatus, and a multiple electron beam exposure apparatus. The writing mechanism **150** includes an electron gun chamber **106**, an electron optical column **102** (electron beam column) and a writing chamber **103**. The electron gun chamber **106** is disposed above the electron optical column **102**. The electron optical column **102** is disposed above the writing chamber **103**.

[0046] In the electron gun chamber **106**, there is disposed an electron gun **201**. In the electron optical column **102**, there is arranged an electron optical system, such as an illumination lens **202**, a shaping aperture array substrate **203**, a blanking aperture array mechanism **204**, a reducing lens **205**, a limiting aperture substrate **206**, an objective lens **207**, a main deflector **208**, and a sub

deflector **209**.

[0047] In the writing chamber **103**, an XY stage **105** is disposed. On the XY stage **105**, there is placed a target object or “sample” **101** such as a mask serving as a writing target substrate when writing (exposure) is performed. The target object **101** is, for example, an exposure mask used in fabricating semiconductor devices, or a semiconductor substrate (silicon wafer) for fabricating semiconductor devices. Furthermore, the target object **101** may be, for example, a mask blank on which resist has been applied and nothing has yet been written. On the XY stage **105**, a mirror **210** for measuring the position of the XY stage **105** is placed.

[0048] The control system circuit **160** includes a control computer **110**, a memory **112**, a high-voltage power supply circuit **120**, a deflection control circuit **130**, digital-analog converter (DAC) amplifier units **132** and **134**, a lens control circuit **136**, a stage control mechanism **138**, a stage position measuring instrument **139**, and a storage device **140** such as a magnetic disk drive. The control computer **110**, the memory **112**, the high-voltage power supply circuit **120**, the deflection control circuit **130**, the lens control circuit **136**, the stage control mechanism **138**, the stage position measuring instrument **139**, and the storage device **140** are connected to each other through a bus (not shown). The DAC amplifier units **132** and **134** and the blanking aperture array mechanism **204** are connected to the deflection control circuit **130**. The sub deflector **209** is composed of at least four electrodes (or “at least four poles”), and controlled by the deflection control circuit **130** through the DAC amplifier unit **132** disposed for each electrode. The main deflector **208** is composed of at least four electrodes (or “at least four poles”), and controlled by the deflection control circuit **130** through the DAC amplifier unit **134** disposed for each electrode. Lenses such as the illumination lens **202**, the reducing lens **205**, and the objective lens **207** are controlled by the lens control circuit **136**. The electron gun **201** is controlled by the high-voltage power supply circuit **120**.

[0049] The high-voltage power supply circuit **120**, the deflection control circuit **130**, the lens control circuit **136**, the stage control mechanism **138**, and the stage position measuring instrument **139** are controlled by the control computer **110**. Information input to the control computer **110** and information calculated in the control computer **110** are stored in the memory **112**.

[0050] Chip data (writing data) on a chip including a plurality of figure patterns is input to the writing apparatus **100** from its outside, and stored in the storage device **140**. Chip data defines information on a plurality of figure patterns configuring a chip pattern. Specifically, for example, coordinates for each vertex are defined for each figure pattern in the order of configuration of the figure. Alternatively, for example, a figure code, coordinates, a size, and the like are defined for each figure pattern.

[0051] The control computer **110** reads the writing data from the storage device **140**, performs multi-stage data processing, and controls, in accordance with the writing sequence, the high-voltage power supply circuit **120**, the deflection control circuit **130**, the lens control circuit **136**, the stage control mechanism **138**, and the stage position measuring instrument **139**.

[0052] Each of the illumination lens **202**, the reducing lens **205**, and the objective lens **207** is configured by an electromagnetic lens, and a reduction optical system is configured by the reducing lens **205** and the objective lens **207**. The illumination lens **202** functions as a collimating lens.

[0053] The position of the XY stage **105** is controlled by the drive of each axis motor (not shown) which is controlled by the stage control mechanism **138**. Based on the principle of laser interferometry, the stage position measurement instrument **139** measures the position of the XY stage **105** by receiving a reflected light from the mirror **210**.

[0054] The electron gun **201** includes a thermionic emission type cathode **10**, a Wehnelt **12** (Wehnelt electrode), an anode **14** (anode electrode), and a limiting aperture substrate **16**. The Wehnelt **12** is arranged between the cathode **10** and the anode **14**. The limiting aperture substrate **16** is arranged at the downstream side of the anode **14** with respect to the advancing direction of an electron beam **200**. Since the Wehnelt **12** and the anode **14** function as electrodes, it goes without

saying that they are made from conductive material.

[0055] FIG. 1 shows configuration elements necessary for describing the first embodiment. Other configuration elements generally necessary for the writing apparatus **100** may also be included therein.

[0056] FIG. 2 is an illustration showing an example of a circuit configuration of an electron gun and a high-voltage power supply circuit according to the first embodiment. In FIG. 2, a passage hole **11** (first opening) through which the electron beam **200** emitted from the cathode **10** passes is formed in the anode **14**. The anode **14** is grounded, and its electric potential is set to be a ground (GND) potential. In other words, the anode **14** is maintained at a relatively positive side potential with respect to the potential of the electron beam **200**. The high-voltage power supply circuit **120** is connected to the electron gun **201**. The high-voltage power supply circuit **120** applies an acceleration voltage between the cathode **10** and the anode **14**. As the acceleration voltage, a potential negative with respect to a ground potential applied to the anode **14** is applied to the cathode **10**. The high-voltage power supply circuit **120** applies a negative bias voltage to the Wehnelt **12**. In other words, the Wehnelt **12** is arranged between the cathode **10** and the anode **14**, and is applied with a relatively negative side potential with respect to the potential of the anode **14**. It will be specifically described.

[0057] In the high-voltage power supply circuit **120**, there are disposed an acceleration voltage power supply **62**, a Wehnelt power supply **64**, and a heater power supply **66**. The cathode (−) side of the acceleration voltage power supply **62** is connected to the cathode **10** through a heater **59** in the electron gun **201**. The anode (+) side of the acceleration voltage power supply **62** is connected to the anode **14** (anode electrode) in the electron gun **201**, and to the ground. An ammeter **70** is connected in series between the anode (+) of the acceleration voltage power supply **62** and the anode **14**. The cathode (−) of the acceleration voltage power supply **62** is branched to be connected to the anode (+) of the Wehnelt power supply **64**. The cathode (−) of the Wehnelt power supply **64** is connected to the Wehnelt **12** (Wehnelt electrode) disposed between the cathode **10** and the anode **14**. The heater power supply **66** is connected to the heater **59**.

[0058] At the time of emission of electron beams, after the arrangement atmosphere in the electron gun **201** is maintained to be in a vacuum state of a predetermined pressure by a vacuum pump (not shown), if the cathode **10** is heated by the heater **59** in a state where a fixed negative Wehnelt voltage (bias voltage) is applied to the Wehnelt **12** from the Wehnelt power supply **64** and a fixed negative acceleration voltage is applied to the cathode **10** from the acceleration voltage power supply **62**, electron beams (electrons) are emitted from the cathode **10**, and the emitted electron beams (electrons) are accelerated, by an acceleration voltage, to travel toward the anode **14**.

[0059] FIG. 3 is a top view showing an example of a configuration of an anode electrode according to the first embodiment. In the examples of FIGS. 2 and 3, the anode **14** includes, for example, a disc electrode substrate **7**. The surface of the electrode substrate **7** is facing the cathode **10**. In other words, the anode **14** includes a surface opposite to the cathode **10**, and the passage hole **11** (first opening) through which the electron beam **200** emitted from the cathode **10** passes is formed in this surface. The passage hole **11** is preferably formed at the center of the electrode substrate **7**. In the anode **14**, at least one opening **15** (second opening) is formed, at a position different from that of the passage hole **11**, in the same surface where the passage hole **11** is formed. For example, at least one opening **15** (second opening) larger than the passage hole **11** is formed.

[0060] Furthermore, as shown in FIGS. 2 and 3, as the at least one passage hole **15**, it is preferable that a plurality of passage holes **15** are formed to be rotationally symmetric about the center of the passage hole **11**.

[0061] The electron beam **200** having passed through the passage hole **11** of the anode **14** travels toward the limiting aperture substrate **16**. A passage hole **17** (third opening) through which the electron beam **200** passes is formed in the limiting aperture substrate **16**, and passage of a portion of the electron beam **200** is limited. For example, a beam spreading wider than a desired radiation

angle is blocked. Furthermore, scattered electrons, etc. are blocked.

[0062] In the case of FIG. 2, the anode **14** includes a supporting cylinder **8** extending to the downstream side with respect to the advancing direction of the electron beam **200** from the outer peripheral part of the electrode substrate **7**. The anode **14** is supported above the limiting aperture substrate **16** through the supporting cylinder **8**. Preferably, the limiting aperture substrate **16** is made from conductive material. In that case, a ground potential is applied to the limiting aperture substrate **16** through the anode **14**.

[0063] FIG. 4 is an illustration showing an example of an internal configuration of an electron gun according to a comparative example of the first embodiment.

[0064] FIG. 5 is an illustration showing an example of an internal configuration of an electron gun according to the first embodiment.

[0065] In FIG. 4, an electron beam **500** emitted from a cathode **510** is accelerated by an acceleration voltage and a bias voltage applied to a Wehnelt **512**, and passes through a passage hole **511** of an anode **514**. Then, it travels to a limiting aperture substrate **516**. In FIG. 4, a crossover formed in the vicinity of the Wehnelt **512** is not shown. The same applies to each following figure. In FIG. 4, a portion of the electron beam **500** having passed through the anode **514** collides with the limiting aperture substrate **516** to emit a secondary electron or a reflected electron. Generally, by performing baking treatment to the electron gun chamber, gas generation from the chamber is promoted, the inside is highly vacuumized, and the number of gas molecules being ion seeds is reduced as small as possible. However, a secondary electron or a reflected electron collides with a gas molecule remaining without being exhausted, thereby generating a cation **521**. The cation **521** generated under the anode **514** is attracted by space charge, passes through the passage hole **511**, and then, collides with the cathode **510** along the electron beam **500**. Thereby, crystals configuring the cathode **510** are damaged to locally form a hollow. This poses a problem of incapable of acquiring a sufficient current distribution. If damaged, the cathode **510** needs to be exchanged.

[0066] Also, in the first embodiment shown in FIG. 5, similarly to the comparative example, the electron beam **200** emitted from the cathode **10** is accelerated by an acceleration voltage and a bias voltage applied to the Wehnelt **12**, and passes through the passage hole **11** of the anode **14**. Then, it travels to the limiting aperture substrate **16**. A portion of the electron beam **200** having passed through the anode **14** collides with the limiting aperture substrate **16** to emit a secondary electron or a reflected electron. Therefore, also in the first embodiment, similarly to the comparative example, by performing baking treatment to the electron gun chamber **106**, gas generation from the chamber is promoted, a secondary electron or a reflected electron collides with a gas molecule remaining without being exhausted, thereby generating a cation **21**. However, according to the first embodiment, as shown in FIG. 5, at least one opening **15** having a different size from the passage hole **11** is formed in the vicinity of the passage hole **11** of the anode **14**. For example, at least one opening **15** larger than the passage hole **11** is formed. Although the opening **15** is desirably circular, for example, it is not limited thereto. Any shape is acceptable as long as a larger area than the area of the passage hole **11** is obtained. For example, the opening **15** may be an ellipse or a rectangle. Alternatively, it may be an arc, for example.

[0067] Since the opening **15** is larger in size than the passage hole **11**, the conductance increases as the area of the opening increases, and the probability of passage of cations also increases. For this reason, it is possible to let many cations **21** go into the electron gun chamber **106** outside the anode **14** from the opening **15** formed on a different position from the electron beam trajectory.

[0068] As shown in FIG. 5, the cathode **10** and Wehnelt **12** are maintained at a relatively negative potential with respect to the anode **14**, and the anode **14** is maintained at a relatively positive potential. Therefore, an electric field is generated between them. Then, by forming the opening **15** in the surface opposite to the cathode **10**, an electric field E leaks also in the opening **15**. Thereby, the cation **21** is drawn by the electric field E . If the size of the opening **15** is smaller than the passage hole **11**, the cation **21** is drawn by the passage hole **11**. According to the first embodiment,

since the size of the opening **15** is larger than the passage hole **11**, in addition to the improvement of the conductance described above, more cations **21** can be drawn by the opening **15** due to the pulling effect of the electric field. For this reason, it is possible to let more cations **21** go into the electron gun chamber **106** outside the anode **14** from the opening **15** formed on a different position from the electron beam trajectory. Thus, the number of cations which collide with the cathode **10** through the passage hole **11** can be suppressed or reduced. In the case of making the size of the opening **15** equal to or smaller than the passage hole **11**, it is preferable to form a plurality of openings **15** and make the total area of the openings larger than the passage hole **11**.

[0069] Furthermore, according to the first embodiment, since the electrode substrate **7** of the anode **14** is arranged to face the cathode **10** and no equipotential structure is arranged on the electrode substrate **7**, the risk of electric discharge can be avoided or reduced.

[0070] Furthermore, as shown in FIG. **3**, by arranging a plurality of openings **15** to be rotationally symmetric (even arrangement), deviation of distribution of emitted cations is hard to occur, and the cations **21** can be efficiently emitted from the backside surface of the anode **14**.

[0071] Next, optimization of the opening **15** is described.

[0072] FIG. **6** is an illustration showing an example of a result of a simulation where cations flow backwards according to a comparative example of the first embodiment.

[0073] FIG. **7** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening at a position away from the electron beam passage hole by a distance y_1 according to the first embodiment.

[0074] FIG. **8** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening at a position away from the electron beam passage hole by a distance y_2 according to the first embodiment.

[0075] FIG. **9** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening at a position away from the electron beam passage hole by a distance y_3 according to the first embodiment.

[0076] In FIGS. **7** to **9**, the relation is $y_3 > y_2 > y_1$, and $y_1 = 3$ mm, $y_2 = 5$ mm, and $y_3 = 10$ mm, for example. Furthermore, each of FIGS. **7** to **9** shows the distance of 15 mm between the anode and the Wehnelt, for example. In the comparative example, as shown in FIG. **6**, many cations flow backwards toward the cathode from the electron beam passage hole. In contrast, it turns out that, by forming the opening **15** at the position away by the distance y_1 , the emission amount of cations from the opening **15** is increased and the emission amount of cations from the electron beam passage hole is decreased. Moreover, it turns out that, by forming the opening **15** at the position away by the distance y_2 , the emission amount of cations from the opening **15** is further increased and the emission amount of cations from the electron beam passage hole is further decreased. However, in the case of forming the opening **15** at the position away by the distance y_3 , it turns out that the emission amount of cations from the opening **15** is less than the case of the distance y_2 , but the emission amount of cations from the electron beam passage hole **11** increases more than the cases of the distances y_1 and y_2 although the emission amount of cations from the electron beam passage hole is less than the comparative example. Thus, it turns out there exists an optimal solution for the distance from the electron beam passage hole **11** to the opening **15**.

[0077] FIG. **10** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening with a diameter size D_{21} .

[0078] FIG. **11** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening with a diameter size D_{22} .

[0079] FIG. **12** is an illustration showing an example of a result of a simulation where cations flow backwards in the case of forming an opening with a diameter size D_{23} .

[0080] In FIGS. **10** to **12**, the relation is $D_{21} < D_{22} < D_{23}$. Each diameter size is larger than the electron beam passage hole **11**. FIG. **10** shows the case where the diameter size D_{21} is larger by 1 mm than the diameter size D_1 of the electron beam passage hole **11**. FIG. **11** shows the case where

the diameter size D22 is larger by 2 mm than the diameter size D1 of the electron beam passage hole 11. FIG. 12 shows the case where the diameter size D23 is larger by 3 mm than the diameter size D1 of the electron beam passage hole 11. As shown in the examples of FIGS. 10 to 12, it turns out that the influence of the change of the diameter size of the opening 15 on the emission amount of cations from the electron beam passage hole 11 is small.

[0081] FIG. 13 is an illustration showing a positional relationship in an electron gun according to the first embodiment. In FIG. 13, there are shown the distance x between the Wehnelt 12 and the anode 14, and the distance y from the passage hole 11 of the anode 14 to the opening 15. The distance y indicates the distance between the centers. Furthermore, in FIG. 13, there are shown the diameter size D1 of the electron beam passage hole 11, and the diameter size D2 of the opening 15.

[0082] FIG. 14 is a graph showing an example of a relationship between an optimal distance from a passage hole of the anode to an opening and a distance between the anode and the Wehnelt according to the first embodiment. The diameter of the passage hole 11 of the anode is 2 mm. In FIG. 14, the ordinate axis represents the distance y from the passage hole 11 of the anode 14 to the opening 15, and the abscissa axis represents the distance x between the anode and the Wehnelt. As shown in FIG. 14, the optimal distance y from the center of the passage hole 11 to the center of the opening 15 satisfies a quadratic equation which uses, as a parameter, the distance x between the anode 14 and the Wehnelt 12. Specifically, the optimal distance y from the center of the passage hole 11 to the center of the opening 15 satisfies the following equation (1) using, as a parameter, the distance x between the anode 14 and the Wehnelt 12.

$$[00001] \quad y = 0.037x^2 - 1.333x + 14.667 \quad (1)$$

[0083] When the distance x between the anode and the Wehnelt is 15 mm, as shown in FIGS. 7 and 8, with respect to the cases where the distance y is 3 mm and the distance y2 is 5 mm, there is no large effect difference between the both cases. In addition, although not shown, in the case of the distance y being 1 mm, since the emission amount of cations from the electron beam passage hole 11 is large, no good effect is acquired. Based on these results, it turns out that the margin of 2 mm is permissible with regard to the distance y. In other words, +1 mm margin for the optimum value of the distance y can be allowed. Therefore, it is preferable for the distance y from the center of the passage hole 11 to the center of the opening 15 to satisfy the following equation (2) which uses, as a parameter, the distance x between the anode 14 and the Wehnelt 12. mm is used as the unit of both the distances x and y.

$$[00002] \quad y = 0.037x^2 - 1.333x + 14.667 \pm 1.000 \quad (2)$$

[0084] FIG. 15 is a table showing an example of a relationship between an optimal distance from a passage hole of the anode to an opening and a distance between the anode and the Wehnelt according to the first embodiment. FIG. 15 shows a result in the case where the acceleration voltage V0 between the cathode 10 and the anode 14 is $-50 \text{ kV} \leq V0 < 0$. In FIG. 15, based on the result of the graph of FIG. 14, when the distance between the anode and the Wehnelt is 6 to 12 mm, the optimal distance from the passage hole 11 of the anode 14 to the opening 15 is 5 to 10 mm. When the distance between the anode and the Wehnelt is 12 to 15 mm, the optimal distance from the passage hole 11 of the anode 14 to the opening 15 is 3 to 5 mm. If the distance between the anode and the Wehnelt is less than 6 mm, since that distance is too short, the electric field at the central part becomes strong, and therefore, it turns out that the effect of letting cations go out from the opening 15 is small. Even when the acceleration voltage V0 is greater than -50 kV on the negative side, the relationship in FIG. 15 is satisfied.

[0085] FIG. 16 is an illustration showing an example of an internal configuration of an electron gun according to a modified example 1 of the first embodiment. The modified example 1 of the first embodiment shown in FIG. 16 is the same as FIG. 5 except that at least one opening 17 is formed in the side surface of the supporting cylinder 8 of the anode 14. By forming at least one opening 17 in the side surface of the supporting cylinder 8, the exhaust efficiency of gas generated in the

supporting cylinder **8** can be increased when performing exhausting by a vacuum pump (not shown). Consequently, the number of emitted cations **21** itself can be reduced. With regard to at least one opening **17**, it is preferable to evenly arrange a plurality of openings **17**.

[0086] FIG. **17** is an illustration showing an example of an internal configuration of an electron gun according to a modified example 2 of the first embodiment. The modified example 2 shown in FIG. **17** is the same as FIG. **5** except that a return portion **18** (convex portion) is further formed to be extended from the backside of the surface, which faces the cathode **10**, of the anode **14** toward the limiting aperture substrate **16**. The return portion **18** is arranged in the vicinity of the outer periphery of at least one opening **15**. FIG. **17** shows the case where the return portion **18** is tubularly (cylindrically) arranged along the outer periphery of the opening **15**. By arranging the return portion **18** at each opening **15**, it becomes easy to temporarily confine the cations **21** in a space surrounded by the backside of the electrode substrate **7**, the internal side surface of the supporting cylinder **8**, and the limiting aperture substrate **16**. By this, the number of collision times of a cation colliding with the backside of the electrode substrate **7** and the internal side surface of the supporting cylinder **8** can be increased. Thereby, the energy of the cation **21** can be attenuated. Thereby, if, even when the cation **21** flows backwards from the electron beam passage hole **11** to the cathode **10**, the influence on the crystal serving as the cathode **10** can be reduced. Although, due to the arrangement of the return portion **18**, it becomes difficult for the cation **21** to pass through the opening **15**, cations whose number is larger than that of cations passing through the electron beam passage hole **11** can be emitted from the opening **15**. If the length L of the return portion **18** is too long, it becomes difficult for the cations **21** to be emitted/discharged through the opening **15**. Therefore, preferably, the length L of the return portion **18** is equal to or less than the diameter size D2 of the opening **15**.

[0087] FIG. **18** is an illustration showing another example of the return portion at the anode backside according to the modified example 2 of the first embodiment. FIG. **18** shows the case where the tubular (cylindrical) return portion **18** whose diameter size is larger than the outer periphery of the opening **15** is arranged surrounding the opening **15**. Thus, there is no necessity of making the arrangement position of the return portion **18** coincide with the outer periphery of the opening **15**.

[0088] Although, in the examples described above, each return portion **18** is arranged to surround the whole circumference of one of the openings **15**, it is not limited thereto. A part of the circumference of the opening **15** may not be enclosed by the return portion **18**. Alternatively, the return portion **18** may be arranged at only a part of the openings **15** without being arranged at all the openings **15**.

[0089] FIG. **19** is a conceptual diagram showing a configuration of a shaping aperture array substrate according to the first embodiment. As shown in FIG. **19**, holes (openings) **22** of p rows long (length in the y direction) and q columns wide (width in the x direction) ($p \geq 2$, $q \geq 2$) are formed, like a matrix, at a predetermined arrangement pitch in the shaping aperture array substrate **203**. In the case of FIG. **19**, for example, holes (openings) **22** of 24×24 , that is 24 holes in the y direction and 24 holes in the x direction, are formed. The number of holes **22** is not limited thereto. For example, it is also preferable to form the holes **22** of 512×512 . Each of the holes **22** is a rectangle (including square) having the same dimension and shape as each other. Alternatively, each of the holes **22** may be a circle with the same diameter as each other. Multiple beams **20** are formed by letting portions of the electron beam **200** individually pass through a corresponding one of a plurality of holes **22**. In other words, the shaping aperture array substrate **203** forms the multiple beams **20**.

[0090] FIG. **20** is a sectional view showing a configuration of a blanking aperture array mechanism according to the first embodiment. In the blanking aperture array mechanism **204**, as shown in FIG. **20**, a blanking aperture array substrate **31** being a semiconductor substrate made of silicon, etc. is disposed on a support table **33**. In a membrane region **330** at the center of the blanking aperture

array substrate **31**, a plurality of passage holes **25** (openings), through each of which a corresponding one of the multiple beams **20** passes, are formed at positions each corresponding to each hole **22** in the shaping aperture array substrate **203** shown in FIG. **19**. A pair of a control electrode **24** and a counter electrode **26**, (blanker: blanking deflector), is arranged in a manner such that the electrodes **24** and **26** are opposite to each other across a corresponding one of the plurality of the passage holes **25**. A control circuit **41** (logic circuit) which applies a deflection voltage to the control electrode **24** for the passage hole **25** concerned is disposed, inside the blanking aperture array substrate **31**, close to each corresponding passage hole **25**. The counter electrode **26** for each beam is grounded.

[0091] In the control circuit **41**, an amplifier (not shown) (an example of a switching circuit) is arranged. As an example of the amplifier, a CMOS (Complementary MOS) inverter circuit serving as a switching circuit is disposed. With regard to inputs (IN) to the CMOS inverter circuit, either an L (low) potential (e.g., ground potential) lower than a threshold voltage, or an H (high) potential (e.g., 1.5 V) higher than or equal to the threshold voltage is applied as a control signal. According to the first embodiment, in a state where an L potential is applied to the input (IN) of the CMOS inverter circuit, the output (OUT) of the CMOS inverter circuit, which is to be applied to the control circuit **41**, becomes a positive potential (Vdd), and then, a corresponding beam is deflected by an electric field due to a potential difference from the ground potential of the counter electrode **26**, and is controlled to be in a beam OFF condition by being blocked by the limiting aperture substrate **206**. In contrast, in a state (active state) where an H potential is applied to the input (IN) of the CMOS inverter circuit, the output (OUT) of the CMOS inverter circuit becomes a ground potential, and therefore, since there is no potential difference from the ground potential of the counter electrode **26**, a corresponding beam is not deflected, and is controlled to be in a beam ON condition by passing through the limiting aperture substrate **206**. Blanking control is provided by such deflection.

[0092] Next, operations of the writing mechanism **150** will be described. Electron beams emitted from the electron gun **201** are led to the target object **101** by the electron optical system.

Specifically, it operates as follows: The electron beam **200** emitted from the electron gun **201** (emission source) almost perpendicularly (e.g., vertically) illuminates the whole of the shaping aperture array substrate **203** by the illumination lens **202**. A plurality of rectangular holes **22** (openings) are formed in the shaping aperture array substrate **203**. The region including all of the plurality of holes **22** is irradiated with the electron beam **200**. For example, rectangular multiple beams (a plurality of electron beams) **20** are formed by letting portions of the electron beam **200** applied to the positions of the plurality of holes **22** individually pass through a corresponding one of the plurality of holes **22** in the shaping aperture array substrate **203**. The multiple beams **20** individually pass through corresponding blankers of the blanking aperture array mechanism **204**. The blanker provides blanking control such that a corresponding beam individually passing becomes in an ON condition during a set writing time (irradiation time).

[0093] The multiple beams **20** having passed through the blanking aperture array mechanism **204** are reduced by the reducing lens **205**, and travel toward the hole in the center of the limiting aperture substrate **206**. Then, the electron beam which was deflected by the blanker of the blanking aperture array mechanism **204** deviates from the hole in the center of the limiting aperture substrate **206** and is blocked by the limiting aperture substrate **206**. In contrast, the electron beam which was not deflected by the blanker of the blanking aperture array mechanism **204** passes through the hole in the center of the limiting aperture substrate **206** as shown in FIG. **1**. Thus, the limiting aperture substrate **206** blocks each beam which was deflected to be in the OFF state by the blanker of the blanking aperture array mechanism **204**. Then, each beam for one shot of the multiple beams **20** is formed by a beam which has been made during a period from becoming beam ON to becoming beam OFF and has passed through the limiting aperture substrate **206**. The multiple beams **20** having passed through the limiting aperture substrate **206** are focused by the objective lens **207** so

as to be a pattern image of a desired reduction ratio. Then, all of the multiple beams **20** having passed through the limiting aperture substrate **206** are collectively deflected in the same direction by the main deflector **208** and the sub deflector **209** in order to irradiate respective beam irradiation positions on the target object **101**. For example, when the XY stage **105** is continuously moving, tracking control is performed by the main deflector **208** so that the beam irradiation position may follow the movement of the XY stage **105**. Ideally, the multiple beams **20** irradiating at a time are aligned at the pitch obtained by multiplying the arrangement pitch of a plurality of holes **22** in the shaping aperture array substrate **203** by the desired reduction ratio described above.

[0094] FIG. **21** is a conceptual diagram showing an example of writing operations according to the first embodiment. As shown in FIG. **21**, a writing region **30** (bold line) of the target object **101** is virtually divided into a plurality of stripe regions **32** by a predetermined width in the y direction, for example. In the case of FIG. **21**, the writing region **30** of the target object **101** is divided in the y direction, for example, into a plurality of stripe regions **32** by the width size being substantially the same as the design size of an irradiation region **34** (writing field) which can be irradiated by one irradiation with the multiple beams **20**. Now, an example of the writing operation will be described.

[0095] First, the XY stage **105** is moved to make an adjustment such that the irradiation region **34** of the multiple beams **20** is located at the left end, or at a position further left than the left end, of the first stripe region **32**. Then, when performing writing to the first stripe region **32**, the XY stage **105** is moved, for example, in the -x direction, so that the writing may relatively proceed in the x direction. The XY stage **105** is moved, for example, continuously at a constant speed.

[0096] After writing to the first stripe region **32**, the stage position is moved in the -y direction by the width size of the stripe region **32**. Thereby, the stripe region **32** to be written is shifted (displaced) in the y direction by the width size of the stripe region **32**.

[0097] Next, an adjustment is made so that the irradiation region **34** of the multiple beams **20** can be located at the right end, or at a position further right than the right end, of the second stripe region **32**. By moving the XY stage **105**, for example, in the +x direction, the writing relatively proceeds in the -x direction. Thereby, writing is performed to the second stripe region **32**.

Henceforth, by repeating similar operations, writing to all the stripe regions **32** is performed.

[0098] Thus, due to performing writing while alternately changing the writing direction, the stage moving time can be reduced, which results in reducing the writing time. Owing to one shot of multiple beams having been formed by individually passing through the holes **22** in the shaping aperture array substrate **203**, a plurality of shot patterns maximally up to as many as the number of the holes **22** are formed at a time.

[0099] In the case of FIG. **21**, each stripe region **32** is written in order while alternately changing the writing direction, but the writing operation is not limited to this. It is also preferable to perform writing in the same direction when writing each stripe region **32**.

[0100] Although, in the case of FIG. **21**, writing is performed to each pixel once, it is not limited thereto. Multiple writing which performs writing a plurality of times to each pixel is also acceptable by making the XY stage **105** pass each stripe region a plurality of times. In that case, it is also preferable to shift the position of the stripe region according to the number of times of multiple writing. Furthermore, it is also preferable to perform multiple writing by writing the same pixel a plurality of times during one movement of the stage.

[0101] FIG. **22** is an illustration showing an example of an irradiation region of multiple beams and a pixel to be written (writing target pixel) according to the first embodiment. In FIG. **22**, the stripe region **32** is divided into a plurality of mesh regions by the beam size of the multiple beams **20**, for example. Each mesh region serves as a writing target pixel **36** (beam irradiation unit region, irradiation position). The size of the writing target pixel **36** is not limited to the beam size, and may be any size regardless of beam size. For example, it may be 1/n (n being an integer of 1 or more) of the beam size. FIG. **22** shows the case where the writing region of the target object **101** is divided, for example, in the y direction, into a plurality of stripe regions **32** by the width size being

substantially the same as the size of the irradiation region **34** (writing field) which can be irradiated by one irradiation of the multiple beams **20**. The x-direction size of the rectangular, including square, irradiation region **34** can be defined by (the number of x-direction beams)×(beam pitch in the x direction). The y-direction size of the rectangular irradiation region **34** can be defined by (the number of y-direction beams)×(beam pitch in the y direction). FIG. **22** shows the case of multiple beams of 24×24 (rows×columns) being simplified to 8×8 (rows×columns). In the irradiation region **34**, there are shown a plurality of pixels **28** (beam writing positions) which can be irradiated with one shot of the multiple beams **20**. The pitch between adjacent pixels **28** is the beam pitch of the multiple beams. A sub-irradiation region **29** (pitch cell region) is configured by a rectangular, including square, region surrounded by the size of beam pitches in the x and y directions. In the example of FIG. **22**, each sub-irradiation region **29** is composed of 4×4 pixels, for example.

[0102] FIG. **23** is an illustration explaining an example of a multi-beam writing operation according to the first embodiment. FIG. **23** shows the case where the inside of each sub-irradiation region **29** is written with four different beams. Furthermore, the example of FIG. **23** shows a writing operation where the XY stage **105** continuously moves at the speed at which the XY stage **105** moves the distance L of eight beam pitches while a ¼ region, namely the region of 1/(the number of beams used for irradiation), in each sub-irradiation region **29** is written. In the writing operation shown in FIG. **23**, for example, while the XY stage **105** moves the distance L of eight beam pitches, different four pixels in the same sub-irradiation region **29** are written (exposed) by being applied with four shots of the multiple beams **20** at a shot cycle T with shifting the irradiation position (pixel **36**) in order by the sub deflector **209**. In order that the relative position between the irradiation region **34** and the target object **101** may not be shifted by the movement of the XY stage **105** while these four pixels **36** are written (exposed), the irradiation region **34** is made to follow the movement of the XY stage **105** by collective deflection of all of the multiple beams **20** by the main deflector **208**. In other words, a tracking control is performed. After one tracking cycle is completed, tracking is reset to return to the previous (last) tracking starting position. Since writing of the pixels in the first column from the right of each sub-irradiation region **29** has been completed, in the next tracking cycle after resetting the tracking, first, the sub deflector **209** provides deflection such that the beam writing position is adjusted (shifted) to write the second pixel column from the right still not having been written in each sub-irradiation region **29**, for example. By repeating this operation during writing the stripe region **32**, as shown in the lower part of FIG. **21**, the position of the irradiation region **34** of the multiple beams **20** is sequentially moved (shifted), such as the irradiation region **34a**, **34b**, **34c**, . . . **34o**, to perform writing.

[0103] As described above, according to the first embodiment, it is possible to prevent or reduce the backward flowing of the cations **21** generated due to collision of electrons having passed through the passage hole **11** of the anode **14** to the cathode **10**.

[0104] Embodiments have been explained referring to specific examples described above. However, the present invention is not limited to these specific examples.

[0105] While the apparatus configuration, control method, and the like not directly necessary for explaining the present invention are not described, some or all of them can be appropriately selected and used on a case-by-case basis when needed. For example, although description of the configuration of the control unit for controlling the writing apparatus **100** is omitted, it should be understood that some or all of the configuration of the control unit can be selected and used appropriately when necessary.

[0106] Furthermore, any electron gun and electron beam writing apparatus that include elements of the present invention and that can be appropriately modified by those skilled in the art are included within the scope of the present invention.

[0107] Additional advantages and modification will readily occur to those skilled in the art.

Therefore, the invention in its broader aspects is not limited to the specific details and representative embodiments shown and described herein. Accordingly, various modifications may

be made without departing from the spirit or scope of the general inventive concept as defined by the appended claims and their equivalents.

Claims

1. An electron gun comprising: a cathode configured to emit an electron beam; an anode electrode configured to include a surface which faces the cathode, and in which there are formed a first opening for passing the electron beam emitted from the cathode, and at least one second opening at a position different from a position of the first opening, the at least one second opening being formed in a same surface as the first opening, and maintained at a relatively positive potential with respect to a potential of the cathode; a limiting aperture substrate arranged at a downstream side of the anode electrode with respect to an advancing direction of the electron beam, configured to be formed with a third opening for passing the electron beam, and limit passage of a portion of the electron beam; and a Wehnelt electrode arranged between the cathode and the anode electrode, configured to be applied with a relatively negative potential with respect to a potential of the anode electrode.
2. The electron gun according to claim 1, wherein the at least one second opening is larger in size than the first opening.
3. The electron gun according to claim 1, wherein a distance from a center of the first opening to a center of the at least one second opening satisfies a quadratic equation which uses, as a parameter, a distance between the anode electrode and the Wehnelt electrode.
4. The electron gun according to claim 1, wherein a distance y from a center of the first opening to a center of the at least one second opening satisfies an equation of $y=0.037x^2-1.333x+14.667\pm 1.000$ which uses, as a parameter, a distance x between the anode electrode and the Wehnelt electrode.
5. The electron gun according to claim 1, wherein, as the at least one second opening, a plurality of second openings are formed to be rotationally symmetric about a center of the first opening.
6. The electron gun according to claim 1, further comprising: a convex portion formed to be extended from a backside of the surface of the anode electrode toward the limiting aperture substrate, and arranged in a vicinity of an outer periphery of each second opening of the at least one second opening.
7. The electron gun according to claim 6, wherein the convex portion is formed to be tubular along the outer periphery of the each second opening.
8. The electron gun according to claim 1, wherein the anode electrode includes a supporting cylinder extending from an outer peripheral part of the anode electrode toward the limiting aperture substrate.
9. An electron beam writing apparatus comprising: a stage configured to mount thereon a target object; an electron gun configured to include a cathode which emits an electron beam, an anode electrode configured to include a surface which faces the cathode, and in which there are formed a first opening for passing the electron beam emitted from the cathode, and at least one second opening at a position different from a position of the first opening, the at least one second opening being formed in a same surface as the first opening, and maintained at a relatively positive potential with respect to a potential of the cathode, a limiting aperture substrate arranged at a downstream side of the anode electrode with respect to an advancing direction of the electron beam, configured to be formed with a third opening for passing the electron beam, and limit passage of a portion of the electron beam, and a Wehnelt electrode arranged between the cathode and the anode electrode, configured to be applied with a relatively negative potential with respect to a potential of the anode electrode; and an electron optical system configured to lead the electron beam emitted from the electron gun to the target object.

10. The electron beam writing apparatus according to claim 9, wherein the at least one second opening is larger in size than the first opening.
